

Module 02 Introduction to IBM InfoSphere DataStage

IBM InfoSphere Information Server for Data Integration



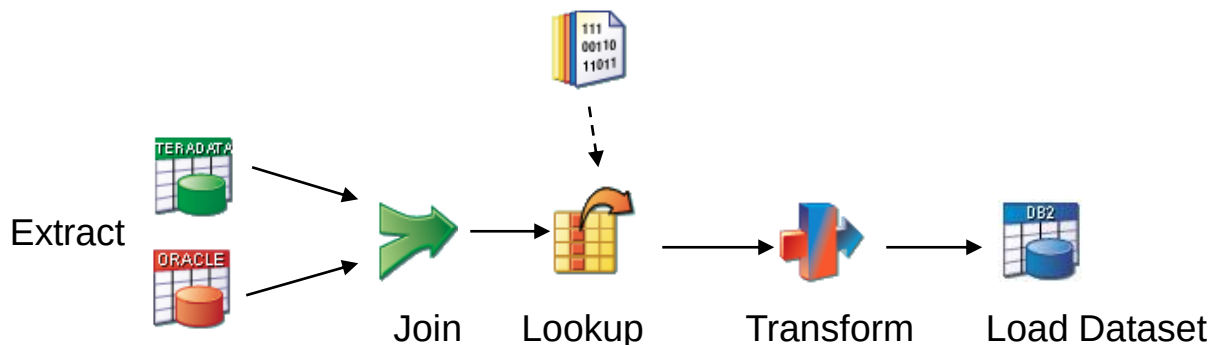
Module Objectives

After completing this topic, you should be able to:

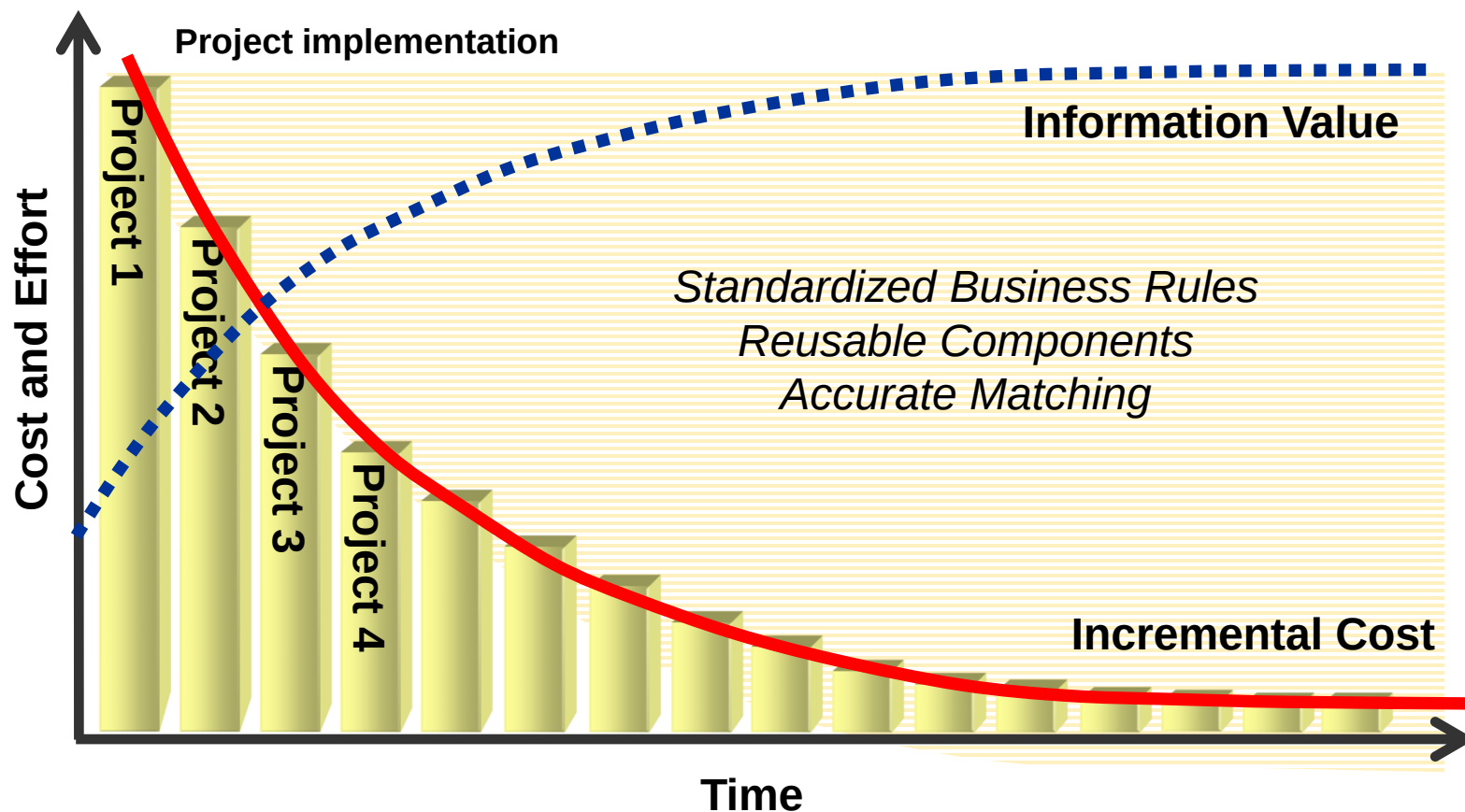
- Explain What IBM InfoSphere DataStage is
- Describe Server/Client Components of DataStage
- Describe Types of Jobs in DataStage

What is IBM InfoSphere DataStage?

- Provides codeless visual design of data flows with hundreds of built-in transformation functions
 - Increase productivity over hand coding
 - Reduce risks of large, complicated integration projects
- Complete Extraction, Transformation, and Loading (ETL) functionality with metadata-driven productivity
- Supports team-based development and collaboration
- Provides integration from across the broadest range of sources
- Supports batch and near real-time operations
- Produces reusable components that can be shared across projects

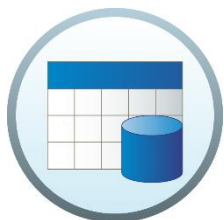


Value of DataStage grows with each project



DataStage Features – Functional Stages

Rich set of functionality packaged into InfoSphere DataStage to do both simple and advanced data integration tasks



■ Feature Rich Components

- Includes dozens of stages and 100+ prebuilt functions addressing common requirements
- Supports looping and caching capabilities to solve most complex data integration tasks



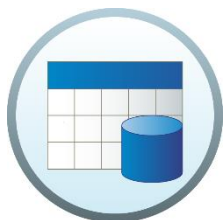
■ Processing Stages

- **Common:** Transformer, Remove Duplicates, Filter, Sort, ...
- **Combining Data:** Join, Funnel, Lookup, Range Lookup, Merge, Aggregator, ...
- **Advanced:** Change Capture
- **DW related:** Slowly Changing Dimension, Surrogate Key Generator
- **Near Real-Time:** Distributed Transaction, MQ, Web Services



DataStage Features – Functional Stages (cont.)

Rich set of functionality packaged into InfoSphere DataStage to do both simple and advanced data integration tasks



■ Files

- Sequential File

- With extensive options supporting fixed width, delimited, variable length record, etc...
- Ability to read and write files in parallel

- Complex Flat File (COBOL)

- FileSets and DataSets (parallel data at rest)

- Big Data File Stage (BDFS) & File Connector

- zOS File

- streaming full records from Mainframe; in conjunction with Classic Federation

■ Extensibility

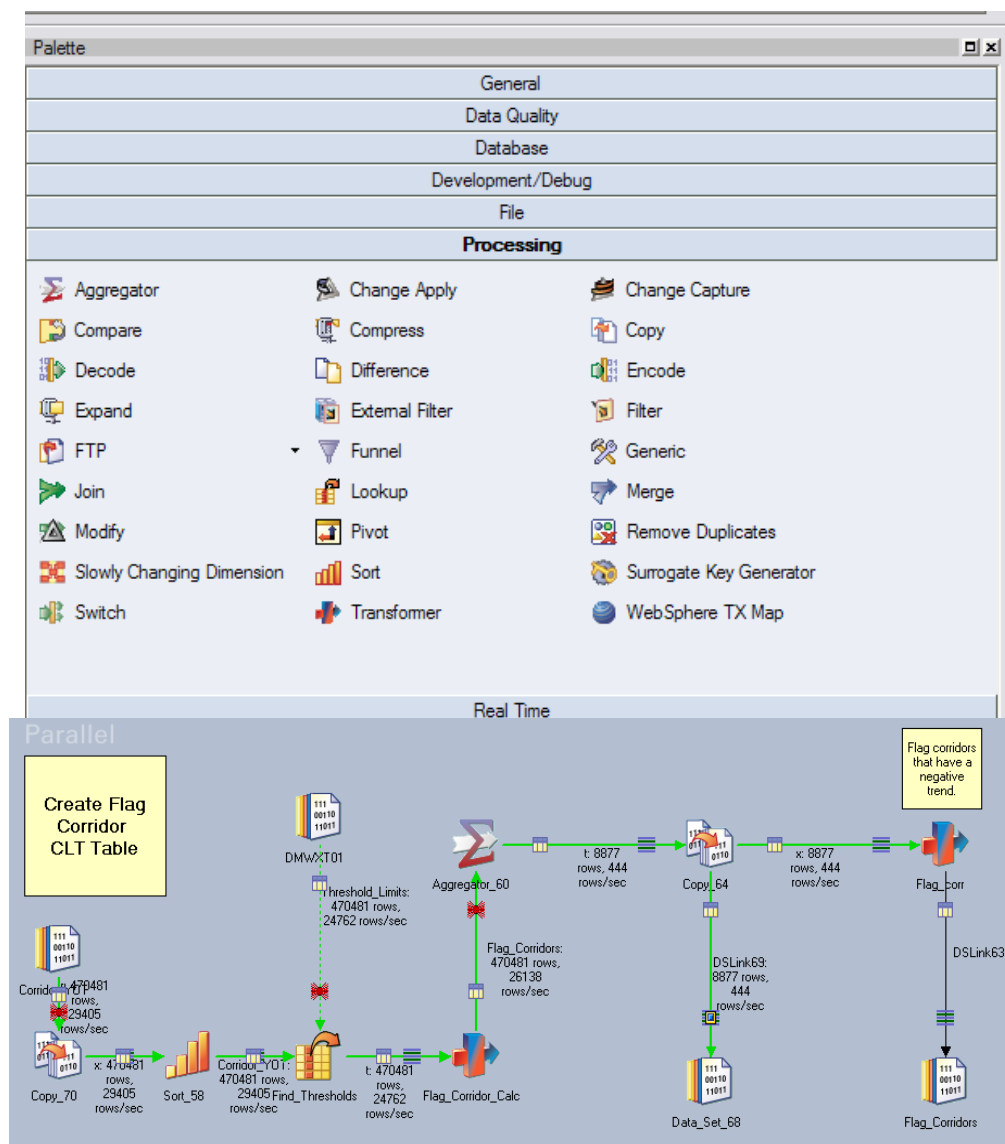
- Java Integration Pack

- Custom routines

- Custom Build, and Wrapped Parallel Stages

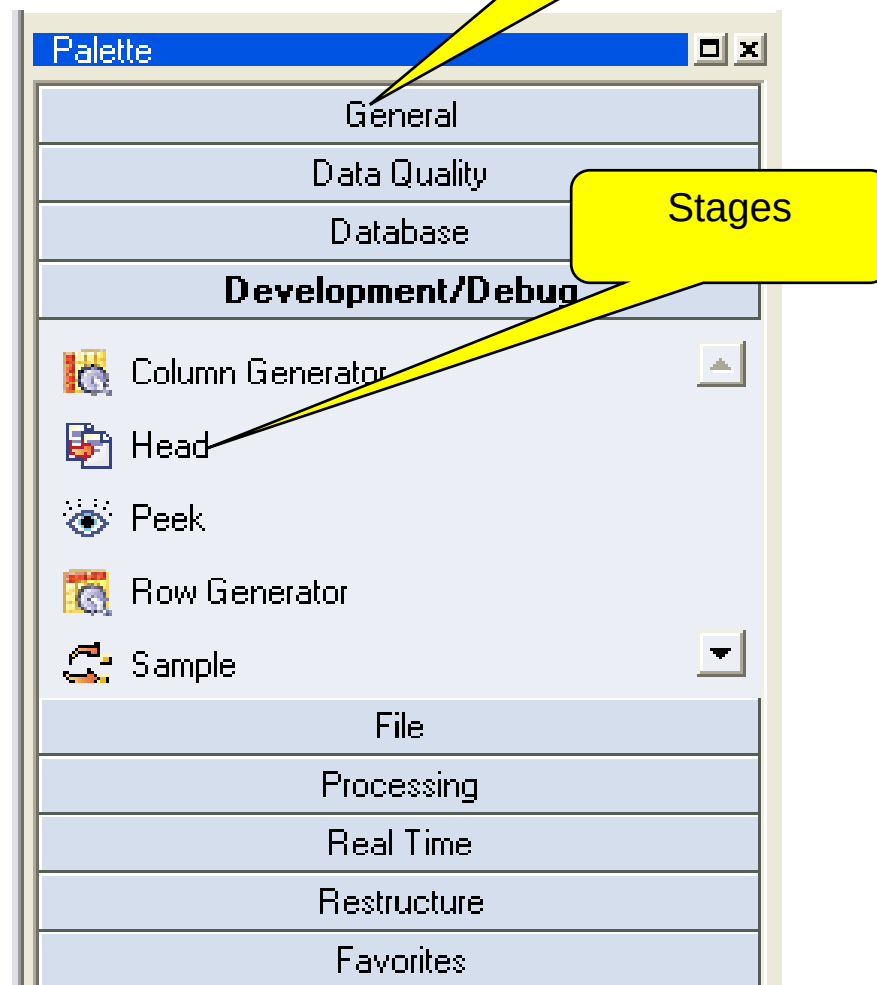
Pre-Built Transformation Stages

- Many pre-built transformation and processing stages for fast job generation and increased productivity
- Large number of built in functions, that make it easy to write advanced transformation logic
- Easy to use stage editors allow quick definition



Pre-Built Transformation stages (cont.)

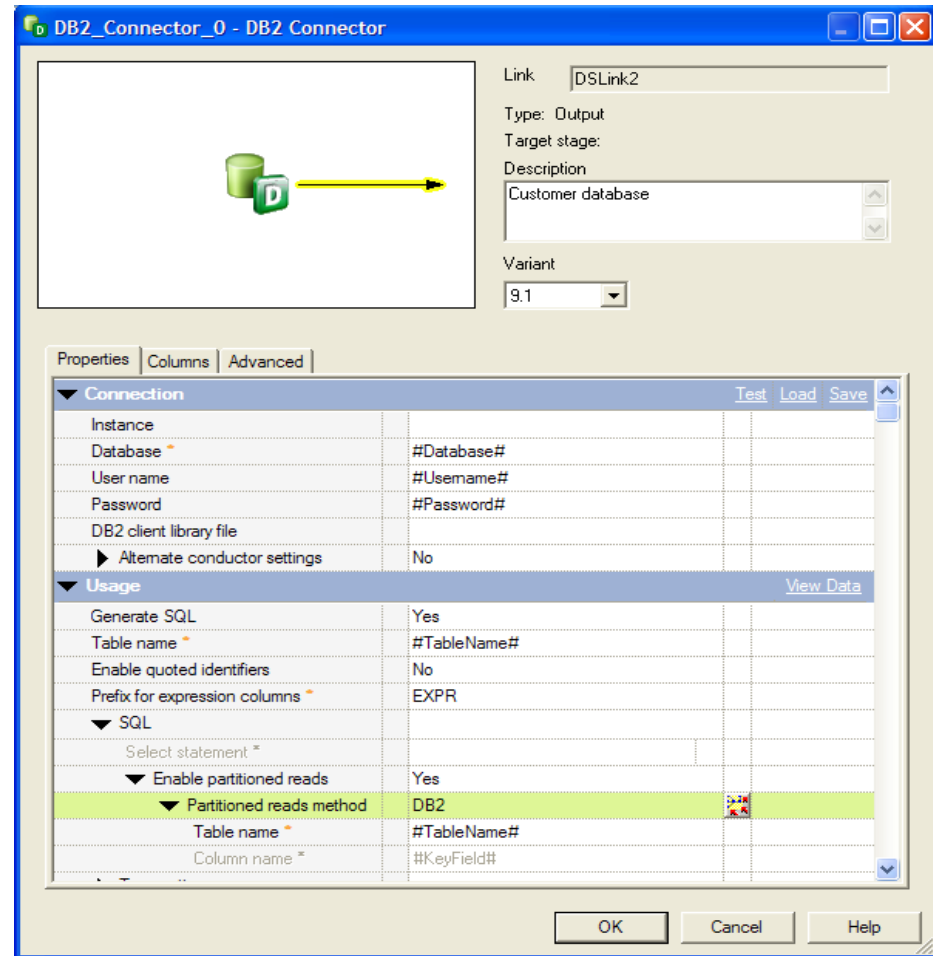
- Stages are organized in categories
 - General
 - Data Quality
 - Stages for data cleansing (QualityStage)
 - Database
 - Stages for connecting to different databases
 - Development
 - Debug
 - File
 - Stages for data files, flat files
 - Processing
 - Real Time
 - Restructure
 - Favorites



DataStage Connectivity options



- DB2 Connector
- ODBC Connector
- Teradata Connector
- Oracle Connector
- Sybase Connector
- Netezza Connector
- and many more.....



DataStage Connectivity options (cont')

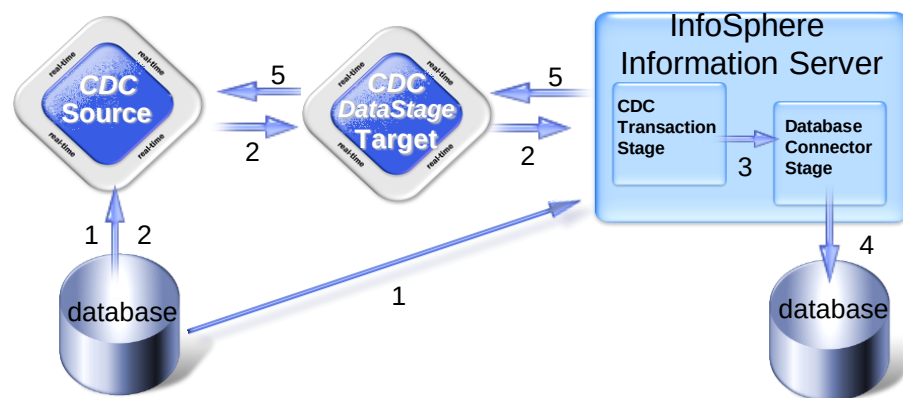
■ Enterprise Applications

- Supporting rich metadata exploration and interchange
- User Interface with application knowledgeable controls
- Native API support



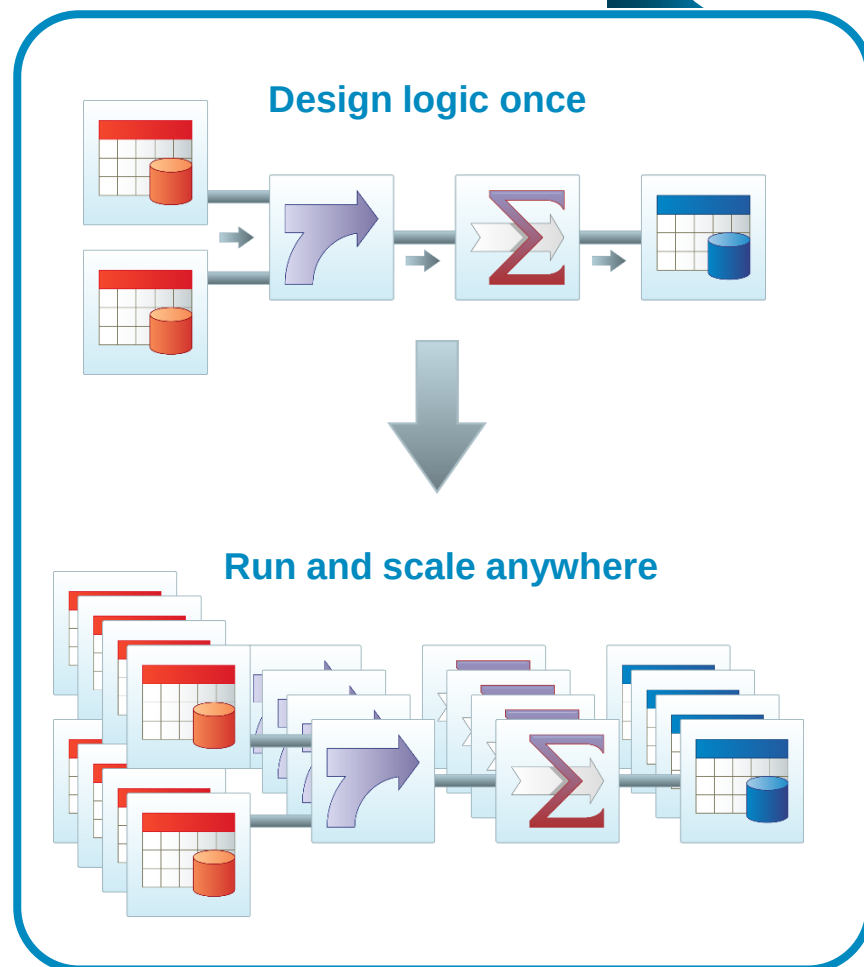
■ Change Data Capture

- Direct streaming of log-based captured InfoSphere CDC data into data integration job
- Bookmarking support for guaranteed delivery of CDC data to target DBMS
- End-to-end metadata support for complete data lineage



Flexible runtime design - scale anywhere

- Push processing to source or target DBs to support ELT, TEL, TETLT, etc... alongside ETL (with Balanced Optimizer)
- Support both batch and real-time processing
- Build complex heterogeneous data integration tasks as web services
- Checkpoint/restart of data integration tasks
- Runtime column propagation allows maximum reuse



DataStage Components – Information Server Web Console

- Web application for administering Information Server
- User interface for:
 - Session management
 - Management of users and groups
 - Logging management
 - Scheduling management

Opening the Web Console

Prior to 11.3 default port is **9080**

After 11.3 already using https with default port is **9445**.

IBM InfoSphere Information Server - Windows Internet Explorer

http://infosrvr:9080/ibm/iis/console/loginForm.jsp?display=...

File Edit View Favorites Tools Help

★ Favorites IBM InfoSphere Information Server

Login

Please enter your information

User name:

Password:

Please note that after some time of inactivity, the system will log you out automatically and ask you to sign in again.

Login

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Information Server
console address

Information
Server
administrator ID

Suite roles - Creating a DataStage User ID

Administration console

IBM InfoSphere Information Server

Home Administration Reporting Repository Management

Help | About | Change Password | Log Out

Navigation

- Contents
- Domain Management
- Session Management
- Users and Groups
 - Users
 - Groups
- Log Management
- Schedule Monitoring

Users

Select Users to Work With

First Name (Given Name) Last Name (Family Name) Filter Clear Filter

Additional Filter Criteria

Item 1-4 out of 4	Last Name	First Name	User Name	Courtesy Title	Business Phone	Location
☐	dsadm	dsadm	dsadm			
☐	dsuser	dsuser	dsuser			
☐	isadmin	isadmin	isadmin			
☐	wasadmin	wasadmin	wasadmin			

Items per Page: 25 50 100

New User
Open User
Add Roles to Multiple Users
Delete

Users

Create new user

Suite roles - Assigning DataStage Roles

IBM InfoSphere Information Server

Home Administration Reporting Repository Management

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Users

Select Users to Work With

Open User

Role and group changes will not affect current users until their next login.

User Name: * dsadm

Password: *

Confirm Password: *

Courtesy Title:

First Name (Given Name): * dsadm

Last Name (Family Name): * dsadm

Instant Messaging ID:

Job Title:

Home Phone Number:

Office Phone Number:

Mobile Phone Number:

Pager Number:

Fax Number:

Email Address:

Business Address:

Roles

Suite

Role	Inherited
☐ Common Metadata Administrator	
☐ Common Metadata Importer	
☐ Common Metadata User	
☐ Suite Administrator	
☒ Suite User	

Suite Component

Role	Inherited
☐ Business Glossary Administrator	
☐ Business Glossary Asset Assigner	
☐ Business Glossary Author	
☐ Business Glossary Basic User	
☐ Business Glossary User	
☒ DataStage and QualityStage Administrator	
☐ DataStage and QualityStage User	
☐ FastTrack Project Administrator	
☐ FastTrack User	
☐ Information Analyzer Data Administrator	
☐ Information Analyzer Project	

Groups

User ID

Assign Suite Administrator role

Assign Suite User role

Assign DataStage User role

Assign DataStage Administrator role

Users

Credential Mapping

- DataStage credentials
 - Required by DataStage in order to log onto a DataStage server
 - Managed by the DataStage Server operating system or LDAP
- Users given DataStage Suite roles in the Web Console do not automatically receive DataStage credentials
 - Users need to be mapped to a user who has DataStage credentials on the DataStage Server machine
 - This DataStage user must have file access permission to the DataStage engine/project files or Administrator rights on the operating system
- Three ways of mapping users (IS ID → OS ID)
 - Many to 1
 - 1 to 1
 - Combination of the above

Credential Mapping (Cont')

IBM InfoSphere Information Server

Home Administration Reporting Repository Management

Help | About | Change Password | Log Out

Navigation

- Contents
- Domain Management
 - Engine Credentials
 - User Registry Configuration
- Session Management
- Users and Groups
 - Users
 - Groups
- Log Management
- Schedule Monitoring

Users Engine Credentials

Select the InfoSphere Information Server engine

Open Configuration "infosrvr"

InfoSphere Information Server Engine Configuration

☐ Share User Registry between InfoSphere Information Server and its engine

You can configure the InfoSphere Information Server and its engine to authenticate users against the same User Registry (e.g. LDAP). If you don't want or can not share the same User Registry, you need to establish credential mapping between the users defined in the registry used by the InfoSphere Information Server and the one used by its engine.

Default Credentials

Define the default credentials below for users that don't have their own credential mapping. Otherwise leave blank to use only mapped credentials.

User Name:
dsadm

Password:
.....

Confirm Password:
.....

Clear Fields

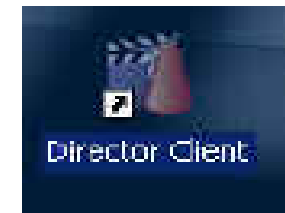
Cancel Save Save and Close

Map DataStage User credential

DataStage Components Clients

- DataStage clients

- Administrator
- Designer
- Director



- DataStage engines

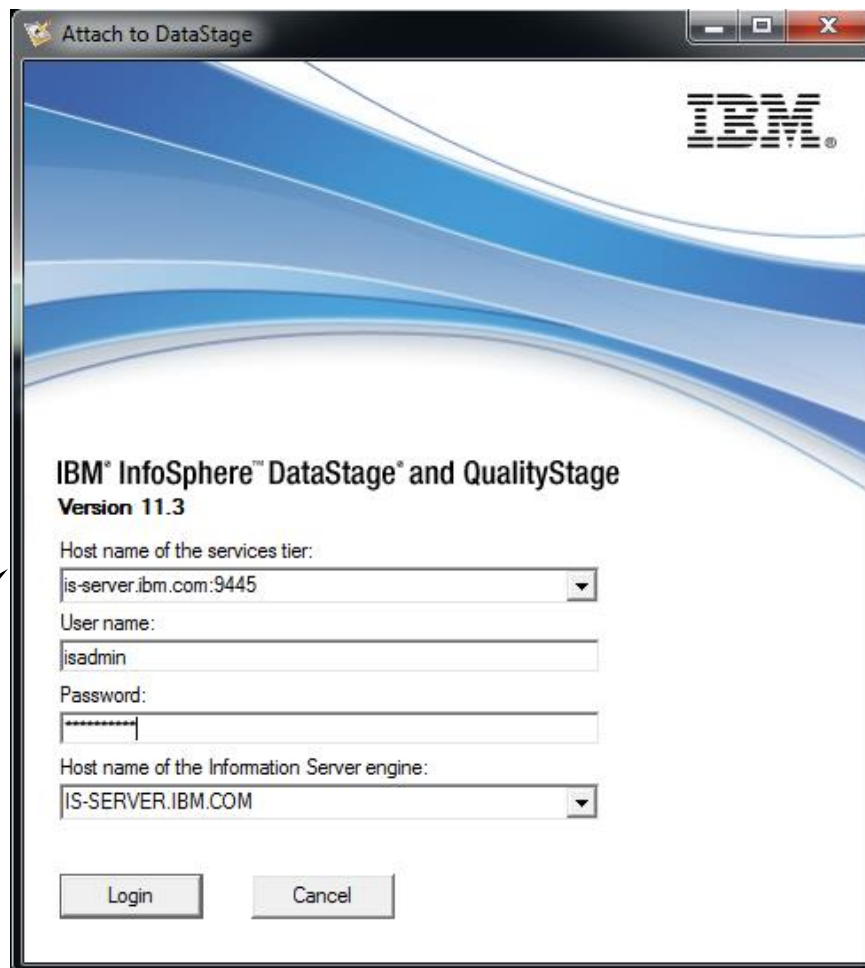
- Parallel engine
 - Runs parallel jobs
- Server engine
 - Runs server jobs
 - Runs job sequences



DataStage Components Administrator Client

Prior to 11.3 default port is **9080**

After 11.3 already using https with default port is **9445**.



Attach to DataStage

IBM®

IBM® InfoSphere™ DataStage® and QualityStage
Version 11.3

Host name of the services tier:
is-server.ibm.com:9445

User name:
isadmin

Password:

Host name of the Information Server engine:
IS-SERVER.IBM.COM

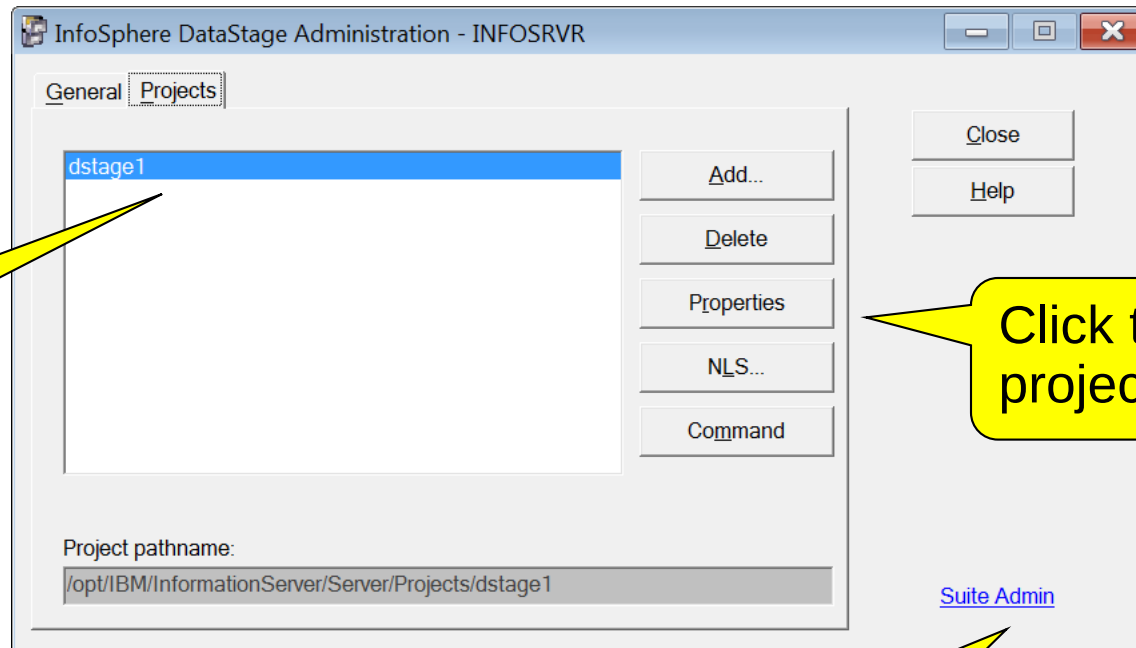
Login Cancel

Host name,
port number
of application
server

DataStage
administrator ID
and password

Name or IP
address of
DataStage server
machine

DataStage Administrator Projects Tab

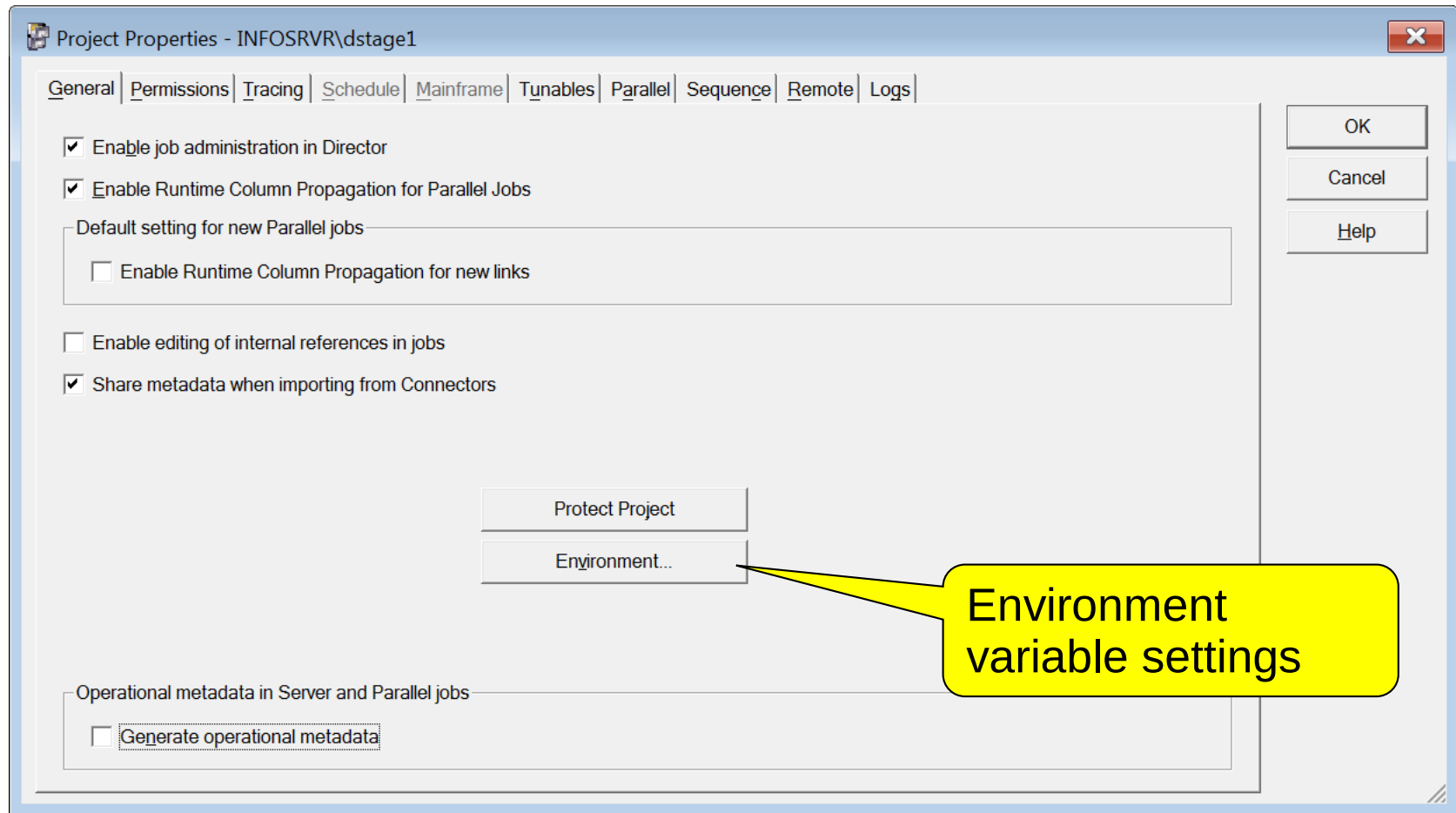


DataStage projects

Click to specify project properties

Link to Information Server Administration console

DataStage Administrator Project Properties



DataStage Designer Client

The screenshot displays the IBM InfoSphere DataStage Designer Client interface. The main window is titled "InfoSphere DataStage Designer - INFOSRVR\dstage1 - [Parallel - LKP_Equality]". The interface includes a menu bar (File, Edit, View, Repository, Diagram, Debug, Import, Export, Tools, Window, Help) and a toolbar. On the left, the "Repository" pane shows a tree structure for "dstage1" with folders for Data Elements, IMS Databases (DBD), IMS Viewsets (PSB/PCB), Jobs, DS-Advanced, shared, Solutions, AggrSales, and CreateSeqJob. Below the repository is a "Palette" with categories like General, Annotation, Data Quality, Database, Development/Debug, File, Processing, Real Time, Restructure, and Favorites. The central "Diagram" pane shows a data flow graph with nodes: "Employee", "Department", "Lookup_Dept", and "EmpDept". The flow is labeled "from_Employee", "Department_lookup", and "to_EmpDept". The bottom pane is the "Log View", which displays a table of events for the job "LKP_Equality".

Log View

Type	Occurred	Event
	2/21/2012 ...	Lookup_Dept,0: Output 0 produced 3 records.
	2/21/2012 ...	Lookup_Dept,1: Input 0 consumed 2 records. ...
	2/21/2012 ...	Lookup_Dept,1: Output 0 produced 2 records.
	2/21/2012 ...	main_program: Step execution finished with status = OK.

Displaying 35 Entries.

Ready

DataStage Designer – Features

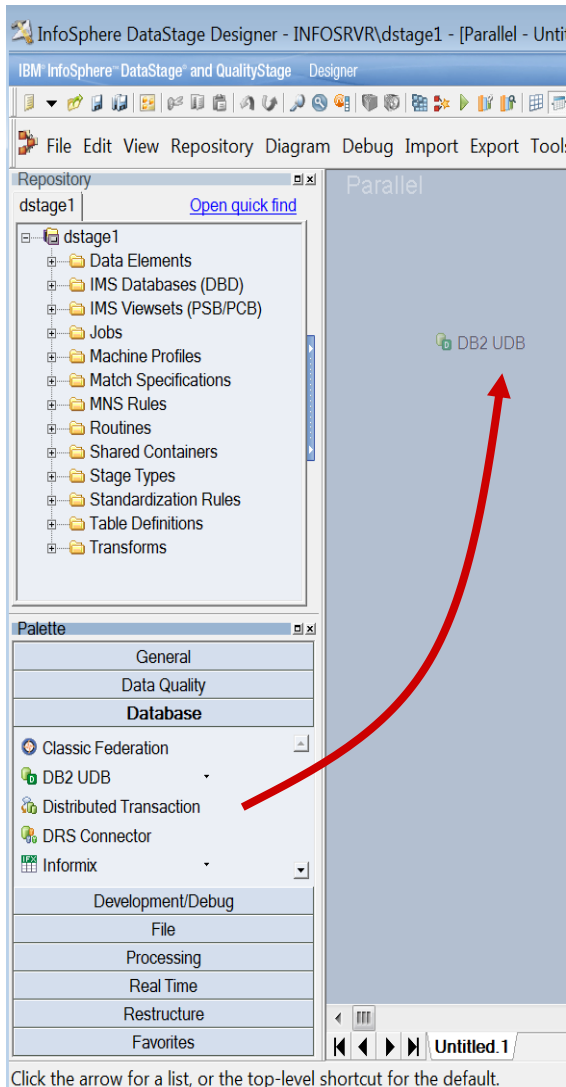
The screenshot shows the IBM InfoSphere DataStage Designer interface. The title bar reads "InfoSphere DataStage Designer - INFOSRVR\dstage1 - [Parallel - LKP_Equality]". The menu bar includes File, Edit, View, Repository, Diagram, Debug, Import, Export, Tools, Window, and Help. The Repository pane on the left shows a tree structure for "dstage1" with folders for Data Elements, IMS Databases (DBD), IMS Viewsets (PSB/PCB), Jobs, DS-Advanced, shared, Solutions, AggrSales, and CreateSeqJob. A red arrow points to the "Open quick find" button. The main canvas displays a data flow diagram with nodes "Employee", "Lookup_Dept", and "EmpDept", connected by arrows labeled "from_Employee" and "to_EmpDept". A red circle highlights the "Parallel" label in the top left of the canvas, with an annotation: "Indicates 'Parallel Canvas' (i.e. DSEE Job)". Another red circle highlights the "Quick Find" button, with an annotation: "Quick Find makes it easy to locate stuff!". A red bracket groups the "General", "Data Quality", "Database", "Development/Debug", "File", "Processing", "Real Time", "Restructure", and "Favorites" categories in the Palette, with an annotation: "These 'boxes' can be docked in various locations within this interface. Just click and drag around...". A red arrow points to the "Customization" menu item, which is open, showing options like "Customize...", "Small icons", "Large icons", "Show text", "Visible groups", "Sort by name", "Clean up", "Load...", "Load project default", "Save as...", "Save as project default", "Reset to factory default", and "Reset to compact default". An annotation points to this menu: "Icons can be made larger by right-clicking inside to access menu." The Log View pane at the bottom shows a table of events:

Type	Occurred	Event
	2/21/2012 ...	Lookup_Dept,0: Output 0 produced
	2/21/2012 ...	Lookup_Dept,1: Input 0 consumed 2 records. ...
	2/21/2012 ...	Lookup_Dept,1: Output 0 produced 2 records.
	2/21/2012 ...	main_program: Step execution finished with status = OK.

The status bar at the bottom indicates "Ready" and "LKP_Equality".

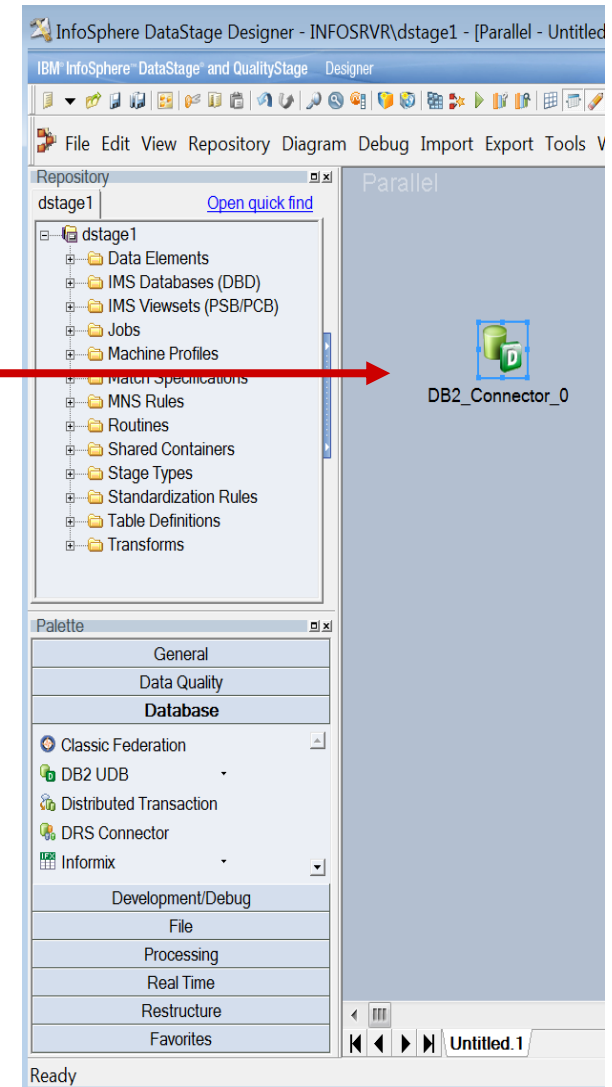
- The DataStage Designer user interface can be customized to your preferences. And it is mainly used for designing your data integration jobs.

DataStage Designer – “Drag & Drop” Design



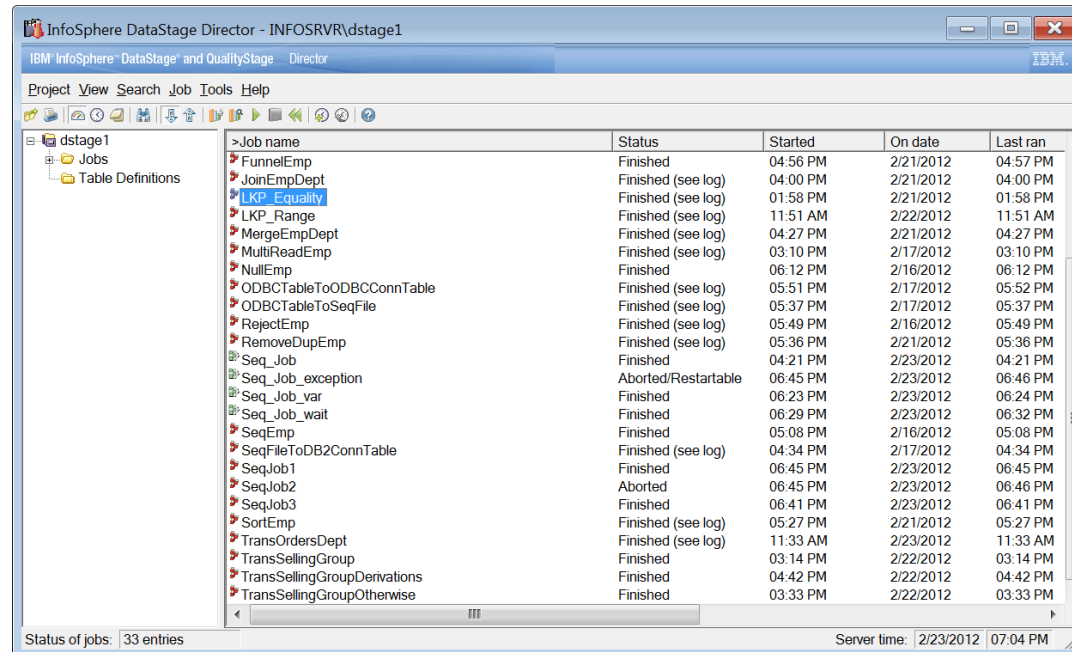
- Left-click and drag the stage(s) onto the canvas.

- You can also left-click on the stage once and then position your mouse cursor on the canvas and left-click again to place the chosen stage there.



DataStage Components - Director Client

Stage/Link name	Link type	Status	Num rows	Started at	Elapsed time	Rows/sec
 Generator_2		Running	5000000	:06:10 PM	00:01:12	69444
 Generator		Running	5000000	:06:10 PM	00:01:12	69444
 Generator_3		Running	5000000	:06:10 PM	00:01:12	69444
 Funnel_1		Running	5000000	:06:10 PM	00:01:12	69444
 DSLink2	<<Pri		5000000			69444
 DSLink7	<<Pri		5000000			69444
 DSLink8	<<Pri		5000000			69444
 DSLink3	>Out		15000000			208333
 Sort		Running	15000000	:06:11 PM	00:01:35	157895
 RemDup		Running	4684681	:06:11 PM	00:01:35	49312
 Switch		Running	1592137	:06:11 PM	00:01:35	16759



Types of Jobs in DataStage

- **Server jobs**
 - Executed by the DataStage server engine
 - **Compiled into Basic**
 - Runtime monitoring in DataStage Director
 - No parallel capability
- **Parallel jobs**
 - Executed by the DataStage parallel engine
 - Built-in functionality for pipeline and partition parallelism
 - **Compiled into OSH (Orchestrate Scripting Language)**
 - OSH operators(Executable C++ class instances)
- **Mainframe jobs**
 - Generate jobs that are compiled and run on a mainframe computer
- **Job sequences (batch jobs, controlling jobs)**
 - Master Server jobs that kick-off jobs and other activities
 - Can kick-off Server or Parallel jobs
 - Runtime monitoring in DataStage Director / Operations console
 - Executed by the Server engine